

Cloud-Based Database (RDS) – Documentation

1. Project Title

AWS RDS Database Deployment Using Custom VPC and EC2

2. Objective

The objective of this project is to deploy and manage a MySQL database using **Amazon RDS** within a custom AWS networking environment. The project demonstrates the creation of networking resources, launching an EC2 instance, connecting to an RDS database, and performing database operations using the MariaDB client.

3. AWS Services Used

- **Amazon VPC**
- **Amazon EC2**
- **Amazon RDS (MySQL)**

4. Project Details

- **VPC Name:** smrthivpc
- **Public Subnet:** smpub
- **Private Subnet:** smpriv
- **Route Tables:**
 - smroute1
 - smroute2
- **Internet Gateway:** SMlg
- **EC2 Instance:** smrdsec2
- **AMI:** Amazon Linux
- **Key Pair:** PPK Key
- **Database Identifier:** database-1
- **Database Engine:** MySQL

5. Project Architecture

- **Custom VPC:** smrthivpc
- **Public Subnet:** smpub
- **Private Subnet:** smpriv
- **Route Tables:** smroute1 and smroute2
- **Internet Gateway:** SMlg
- **EC2 Instance:** smrdsec2
- **Amazon RDS Database:** database-1

6. Steps Followed to Deploy the Database

Step 1: Create a Custom VPC

1. Open the AWS Management Console.
2. Navigate to the VPC service.
3. Create a VPC named **smrthivpc**.
4. Configure the required CIDR block.
5. Save the VPC configuration.

Step 2: Create Subnets

1. Create a public subnet named **smpub**.
2. Create a private subnet named **smpriv**.
3. Associate both subnets with **smrthivpc**.

Step 3: Create an Internet Gateway

1. Create an Internet Gateway named **SMIg**.
2. Attach the Internet Gateway to **smrthivpc**.

Step 4: Create Route Tables

1. Create route table **smroute1**.
2. Create route table **smroute2**.
3. Configure routing for internet access through **SMIg**.
4. Associate the route tables with their respective subnets.

Step 5: Launch an EC2 Instance

1. Navigate to the EC2 Dashboard.
2. Launch a new instance named **smrdsec2**.
3. Select **Amazon Linux AMI**.
4. Select the VPC **smrthivpc**.
5. Choose subnet **smpub**.
6. Create and download a **PPK Key Pair**.
7. Configure the required security group settings.
8. Launch the instance.

Step 6: Create an Amazon RDS Database

1. Navigate to the Amazon RDS Console.
2. Click **Create Database**.
3. Select **MySQL** as the database engine.
4. Choose the desired database template.
5. Enter **database-1** as the database identifier.

6. Configure the master username and password.
7. Select **Connect to an EC2 Compute Resource**.
8. Choose the EC2 instance `smrdsec2`.
9. Review the settings and create the database.
10. Wait until the database status changes to **Available**.

Step 7: Connect to the EC2 Instance

1. Open PuTTY.
2. Load the downloaded PPK key file.
3. Enter the public IP address of `smrdsec2`.
4. Connect to the instance.

Step 8: Install MariaDB Client

1. Update the system packages.
`sudo yum update -y`
2. Install the MariaDB client.
`sudo yum install mariadb105 -y`
3. Verify the installation.
`mysql --version`

Step 9: Connect to the RDS Database

1. Obtain the RDS endpoint from the Amazon RDS Console.
2. Connect to the database using:
`mysql -h -u admin -p`
3. Enter the database password when prompted.

Step 10: Create a Database

1. Create a database named `students`.

`CREATE DATABASE students;`

2. Select the database.

`USE students;`

Step 11: Create a Table

1. Create a table named `mca`.
1. `CREATE TABLE mca (`
2. `name VARCHAR(50),`
3. `usn INT`

);

Step 12: Perform CRUD Operations

Insert Records

4. INSERT INTO mca VALUES ('Smrithi', 163);

INSERT INTO mca VALUES ('Megha', 194);

View Records

SELECT * FROM mca;

Output:

Name	USN
Smrithi	163
Megha	194

Update Record

5. UPDATE mca

6. SET usn = 165

WHERE name = 'Smrithi';

View Updated Record

SELECT * FROM mca;

Delete Record

7. DELETE FROM mca

WHERE name = 'Megha';

View Final Table

SELECT * FROM mca;

Output

- Created database **students**.
- Created table **mca** with fields **name** and **usn**.
- Inserted records for **Smrithi** and **Megha**.
- Retrieved records using SELECT statements.
- Updated existing records successfully.
- Deleted records successfully.
- Verified database operations using CRUD commands.

9. Conclusion

This project demonstrates the deployment of a **MySQL Database** using **Amazon RDS** within a custom AWS networking environment. A custom VPC (**smrithivpc**) was configured with public and private subnets, route tables, and an internet gateway. An **Amazon Linux EC2 Instance** (**smrdsec2**) was used to connect to the **RDS Database** (**database-1**) using the **MariaDB Client**, and CRUD operations were successfully performed to verify database connectivity and functionality.